Pitchika Vivek Gautham

ch24btech11028@iith.ac.in | +91 8367295335 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.97
NARAYANA IIT ACADEMY	Class XII (AP -SSC)	2024	98.10%
NARAYANA E.M SCHOOL	Class X (AP-SSC)	2022	98.10%

PROJECTS

REGULATION INTELLIGENCE RAG -- Document Q&A with Evaluation Pipeline*Personal Project | Python, Flask, Qdrant, sentence-transformers, Gemini*

- Developed a **RAG Q&A system** over FIA 2026 F1 regulation docs (**~690 pages**) using **hybrid retrieval** (dense + BM25 + RRF) and **cross-encoder reranking** -- **95% hit-rate@5** with page-level citations.
- Designed a machine-verified eval pipeline (**119 questions**) measuring hit-rate@5, MRR, and LLM-judge faithfulness, plus a paraphrase-robustness test exposing a **38-point drop** under natural phrasing.
- Shipped a **streaming Flask API** + animated web UI with live PDF ingestion, a quota-aware multi-model LLM fallback, and **Dockerized, CI-tested code**.

Multi-Agent Data Analysis & Forecasting Platform*Personal Project | Python, FastAPI, LangGraph, React, Gemini/Groq*

- Architected an **LLM agent-orchestration system** using the Supervisor + Agents-as-Tools pattern on LangGraph, where one routing LLM delegates to **5 specialist agents** exposed as tools and coordinated through shared graph state.
- Built a **ReAct code agent** that writes and self-corrects pandas in a restricted sandbox, and a forecasting agent that backtests **4 models** (XGBoost, linear regression, etc) before predicting with confidence intervals.
- Developed interactive **Plotly visualizations** (2D/3D) and grounded insight narration with anti-hallucination guarantees, so agents explain results using only pre-computed evidence -- never fabricated numbers.

Fraud Detection Risk-Threshold Optimization

Personal Project | Personal Project |Python, XGBoost, scikit-learn, SHAP

- Balanced **~$809K** in prevented fraud against just **~$24K** in customer-friction cost across recall and precision curves, via a dollar-denominated cost-tradeoff model translating a technical metric into a business call across **1.3M+** transactions.
- Improved fraud recall from **89.7%** to **98.4%** at a fixed friction budget by diagnosing a severe class-imbalance issue (**0.58%** fraud rate) that made standard accuracy misleading, and reframing evaluation around **PR-AUC (0.95** vs. **0.37)**.
- Caught and fixed a data-leakage bug, confirming the **$809K** recommendation held post-correction (**PR-AUC 0.9512 to 0.9488**), then shipped a live interactive dashboard (GitHub Pages) for stakeholders.

POSITIONS OF RESPONSIBILITY

SUGC Representative | IIT Hyderabad

- Represented the academic interests of a **600-student batch** in **quarterly Senate-level deliberations** involving **100+ faculty members** by consolidating student feedback and coordinating with the Academic Affairs Secretary and Academic Affairs Council.
- Elevated batch-level academic concerns to institute governance by structuring student feedback into clear discussion points and presenting them through formal institutional channels.

Class Representative | IIT Hyderabad

- Served as the primary bridge between the student batch and faculty by organising and formally presenting academic feedback, building skills in structured communication, documentation, and batch-level coordination.
- Coordinated timely dissemination of academic updates, schedule changes, and administrative notices to **40+ students**, developing reliability, accountability, and organised information management as core competencies.

Hospitality Coordinator | DIESTA, IIT Hyderabad

- Supported hospitality and event operations for **3,000+ members** across **13 competition rooms**, event grounds, pronight production, and international guest-house accommodation by coordinating with event heads and resolving operational requirements throughout the event.
- Enabled smooth execution across multiple event venues by managing room readiness, accommodation coordination, on-ground hospitality, and live pronight production requirements.

SKILLS

Technical Skills: Python, SQL, C++, LangGraph, LangChain, LangSmith, MCP, FastAPI, RAG, Vector Databases, Agentic Architecture, NLP, Machine Learning, Deep Learning,

Product & Communication Skills: User Research, Stakeholder Management, Structured Communication, Resource Allocation and Trade-off Decision-Making, Real-Time Problem-Solving, Creativity.

RELEVANT COURSES

Computer Science courses: Introduction to Programming, Data Structures and its Applications.

Mathematics courses: Numerical Methods, Linear Algebra, Differential Equations, Calculus I & II.

Machine Learning Courses: Applied Mathematics, Introduction to AI

SCHOLASTIC ACHIEVEMENTS

-**Achieved a high SGPA of 9.4/10.0** in the 3rd semester, demonstrating strong academic and technical performance in coursework.

-**Solved 250+ Data Structures and Algorithms (DSA) problems** across major coding platforms, demonstrating strong algorithmic problem-solving and analytical skills.